of special relativity dictated that any object approaching the speed of light would require an infinite amount of energy to get there. And an object that did somehow manage to travel faster than light could, in theory, travel into its own past and prevent itself from leaving in the first place—a paradox. Light speed is still plenty fast—about 300,000 kilometers per second—but Einstein had discovered that speed was the fastest any object could travel, signal, or influence any other object.

In a follow-up paper the same year, Einstein extended his theory of relativity to modify Newton's laws of motion, discovering in the process his famous equation that shows mass is a form of energy: $E = mc^2$. And these were just two of the papers Einstein published during his "miraculous year" of 1905. He also published two more seminal papers, on the behavior of atoms and the interaction of light and matter—the work for which he later won a Nobel Prize.

In his work on relativity, Einstein was guided, in part, by the work of the Austrian physicist and philosopher Ernst Mach. Mach believed that science should be based on descriptive laws that don't make any claims about the true nature of the world—he dismissed such claims as unnecessary for the practice of science. To Mach, one of the worst offenders was the great god of physics, Isaac Newton himself. Newton's masterwork, the *Principia*, opened with the assumption that space and time were absolute entities unto themselves, with real existence out in the world. This "conceptual monstrosity of absolute space" was, in Mach's view, "purely a thought-thing which cannot be pointed to in experience." Mach thought that a proper science of mechanics would dispense with these kinds of ontological claims—claims about what things actually exist in the real world—and instead simply lay down descriptive, mathematical laws that accurately predict the observed motion of all objects. Good theories, according to Mach, were about connecting observations, not about positing things that couldn't be observed at all.

The laws of thermodynamics, developed in the early 1800s, were the very model of a modern physical theory, according to Mach. As laid down by Carnot, Joule, and others, thermodynamics simply quantified the observable behavior of heat in steam engines and elsewhere in the world, allowing prediction without positing any extraneous unobservable ideas about the nature of heat itself. Thermodynamics didn't rely on abstruse,